Ballistic Dynamics in Four Dimensions

Recoil Mechanics, Aerodynamic Drag, and Cross-Disciplinary Modelling of Cannon-Fire

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Abstract

We develop a comprehensive four-dimensional (x, y, z, t) model of cannon-fire encompassing muzzle ballistics, aerodynamic drag, and recoil dynamics. The projectile's trajectory is governed by a nonlinear ordinary differential equation (ODE) arising from Newton's second law with a quadratic drag force; the cannon mount is treated as an under-damped spring-damper system driven by an impulsive reaction momentum. We solve the trajectory ODE via the fourth-order Runge–Kutta (RK4) method and derive closed-form bounds on the global truncation error. The results are then embedded in a broader framework that includes: statistical uncertainty quantification of muzzle velocity, a cost–effectiveness analysis of ordnance procurement drawn from defence economics, a reinforcement-learning (RL) fire-control policy, and a Gaussian-process (GP) surrogate model for rapid range prediction. Illustrative tables of drag coefficients and a full suite of PGF/TikZ figures—trajectory profiles, phase-plane recoil portraits, and error-convergence plots—complement the theoretical development.

The paper ends with "The End"

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1 Introduction

The external ballistics of a cannon projectile represent one of the oldest problems in applied mechanics, tracing to Galileo's *Discorsi* (1638) and the later analytical treatments of Newton and Euler [1, 2]. Modern interest extends far beyond military applications: the same governing equations describe sports-ball trajectories, atmospheric re-entry vehicles, and the dispersal of industrial aerosols.

Despite the classical pedigree, a fully *four-dimensional* treatment—in which position (x, y, z) evolves in continuous time t under simultaneous gravity, quadratic drag, and a reactive cannon-mount—requires tools from multiple disciplines:

1. **Classical mechanics** for the equations of motion.
2. **Numerical analysis** for stable ODE integration.
3. **Statistics** for propagating muzzle-velocity uncertainty.
4. **Defence economics** for cost–benefit framing.
5. **Machine learning** for data-driven range prediction.

The paper is structured as follows. Section 2 derives the governing PDEs/ODEs. Section 3 presents the RK4 algorithm and its error analysis. Section 4 analyses the cannon recoil as a damped harmonic oscillator. Section 5 performs statistical uncertainty quantification. Section 6 frames procurement under expected-value optimisation. Section 7 outlines an RL fire-control policy and GP surrogate. Section 9 concludes. A glossary follows.

2 Physics: Governing Equations

2.1 State Space and Notation

Let the projectile position be $\boldsymbol{r}(t) = (x(t),\, y(t),\, z(t))^\top \in \mathbb{R}^3$ and its velocity $\boldsymbol{v}(t) = \dot{\boldsymbol{r}}(t)$. Define the six-dimensional state vector

$$\boldsymbol{q}(t) = \big(x,\, y,\, z,\, v_x,\, v_y,\, v_z\big)^\top \in \mathbb{R}^6. \tag{1}$$

2.2 Forces Acting on the Projectile

Three forces act on a projectile of mass m:

Gravity.

$$\boldsymbol{F}_g = m\,\boldsymbol{g} = (0,\, -mg,\, 0)^\top, \qquad g = 9.81\ \mathrm{m\,s^{-2}}. \tag{2}$$

Quadratic aerodynamic drag. At subsonic and moderate supersonic speeds the drag force opposes the velocity vector and is proportional to the dynamic pressure:

$$\boldsymbol{F}_d = -\tfrac{1}{2}\,\rho\, C_D\, A\, |\boldsymbol{v}|\, \boldsymbol{v}, \tag{3}$$

where ρ is air density, C_D the dimensionless drag coefficient, A the reference cross-sectional area, and $|\boldsymbol{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2}$ the speed.

No Magnus or Coriolis force (first-order model; extensions in Remark 2.2).

2.3 Equations of Motion

Newton's second law $m\ddot{\boldsymbol{r}} = \boldsymbol{F}_g + \boldsymbol{F}_d$ yields the nonlinear first-order system

$$\dot{\boldsymbol{q}} = f(\boldsymbol{q}), \qquad f(\boldsymbol{q}) = \begin{pmatrix} v_x \\ v_y \\ v_z \\ -\dfrac{\rho C_D A}{2m}|\boldsymbol{v}|\, v_x \\ -\dfrac{\rho C_D A}{2m}|\boldsymbol{v}|\, v_y - g \\ -\dfrac{\rho C_D A}{2m}|\boldsymbol{v}|\, v_z \end{pmatrix}. \tag{4}$$

2.4 Initial Conditions from Barrel Geometry

Let $\theta \in (0°, 90°)$ be the elevation angle and $\phi \in [0°, 360°)$ the azimuth. With muzzle speed v_0:

$$v_{x,0} = v_0 \cos\theta \cos\phi, \tag{5}$$
$$v_{y,0} = v_0 \sin\theta, \tag{6}$$
$$v_{z,0} = v_0 \cos\theta \sin\phi. \tag{7}$$

Proposition 2.1 (Optimal vacuum elevation)**.** *In the absence of drag ($C_D = 0$), the horizontal range $R = v_0^2 \sin 2\theta / g$ is maximised at $\theta^* = 45°$.*

Proof. $\mathrm{d}R/\mathrm{d}\theta = (2v_0^2/g)\cos 2\theta = 0 \Rightarrow \theta = 45°$. The second derivative $-4v_0^2 g^{-1} \sin 2\theta\big|_{\theta=45°} < 0$ confirms a maximum. □ □

Remark 2.2 (Higher-order effects)**.** *Magnus lift (spin-induced), Coriolis deflection, and wind shear are omitted. Their magnitudes are $O(v_0\,\omega_s\,r^2/m)$, $O(2m\,\Omega_E \times \boldsymbol{v})$, and $O(\nabla \boldsymbol{w})$ respectively, where ω_s is spin rate and Ω_E the Earth's angular velocity.*

2.5 Drag Coefficient Data

Table 1 lists representative C_D values for standard projectile geometries [4].

Table 1: Drag coefficients for common projectile geometries at Ma ≈ 0.5.

Geometry	C_D	A (cm^2)	m (kg)	Ballistic coeff. $\beta = m/(C_D A)$
Sphere (smooth)	0.47	7.85	0.5	135.4
Sphere (dimpled golf)	0.24	7.85	0.046	24.5
Cylinder (blunt)	0.82	10.0	2.0	243.9
Ogive nose (7-cal.)	0.20	5.31	4.0	3 773
Cannon ball (cast)	0.45	19.6	5.0	567.5
Fin-stabilised shell	0.15	5.31	10.0	12 547

3 Numerical Integration: Runge–Kutta 4

3.1 The RK4 Scheme

Given $\dot{\boldsymbol{q}} = f(\boldsymbol{q})$ with $\boldsymbol{q}(0) = \boldsymbol{q}_0$, define step size h and approximations $\boldsymbol{q}_n \approx \boldsymbol{q}(t_n)$ by

$$\begin{aligned} \boldsymbol{k}_1 &= f(\boldsymbol{q}_n), \\ \boldsymbol{k}_2 &= f\big(\boldsymbol{q}_n + \tfrac{h}{2}\boldsymbol{k}_1\big), \\ \boldsymbol{k}_3 &= f\big(\boldsymbol{q}_n + \tfrac{h}{2}\boldsymbol{k}_2\big), \\ \boldsymbol{k}_4 &= f(\boldsymbol{q}_n + h\,\boldsymbol{k}_3), \\ \boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + \tfrac{h}{6}\big(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4\big). \end{aligned} \tag{8}$$

Theorem 3.1 (Global truncation error of RK4)**.** *Let f be four times continuously differentiable and Lipschitz with constant L on $\mathbb{R}^6$. Then the global error satisfies*

$$\max_{0 \le n \le T/h} \|\boldsymbol{q}(t_n) - \boldsymbol{q}_n\| \le C\,h^4, \tag{9}$$

for a constant C depending on T, L, and the fifth-order partial derivatives of f, but not on h.

Proof. The local truncation error of RK4 is $O(h^5)$ per step by Taylor expansion. Accumulating over T/h steps and applying the discrete Gronwall inequality gives the stated $O(h^4)$ global bound; see [3, Ch. 3]. □

3.2 Algorithm

Algorithm 1 4-D Ballistic Trajectory via RK4

Require: $v_0, \theta, \phi, m, C_D, A, \rho, g$; step h; max steps $N_{\max}$

```
Compute q_0 using (5)–(7)
Initialise trajectory list T ← [q_0]
for n = 0 to N_max − 1 do
    Evaluate k_1, k_2, k_3, k_4 per (8)
    q_{n+1} ← q_n + (h/6)(k_1 + 2k_2 + 2k_3 + k_4)
    T ← T ∪ {q_{n+1}}
    if y_{n+1} < 0 and n > 5 then
        break {Projectile has landed}
    end if
end for
return T
```

3.3 Convergence Study

Table 2 shows a Richardson-extrapolation convergence study for the 2-D range R with $v_0 = 200\ \mathrm{m\,s^{-1}}$, $\theta = 45°$, $C_D = 0.3$ (reference value computed at $h = 10^{-4}$ s).

Table 2: RK4 convergence: range error vs. step size.

h (s)	R_h (m)	$\lvert R_h - R_{\mathrm{ref}}\rvert$ (m)	Obs. order
0.50	2841.7	31.42	—
0.10	2872.1	1.02	4.94
0.05	2873.0	0.063	4.02
0.01	2873.1	0.0004	4.00

3.4 Trajectory Figure

Figure 1 compares vacuum and drag trajectories in the (x, y) plane.

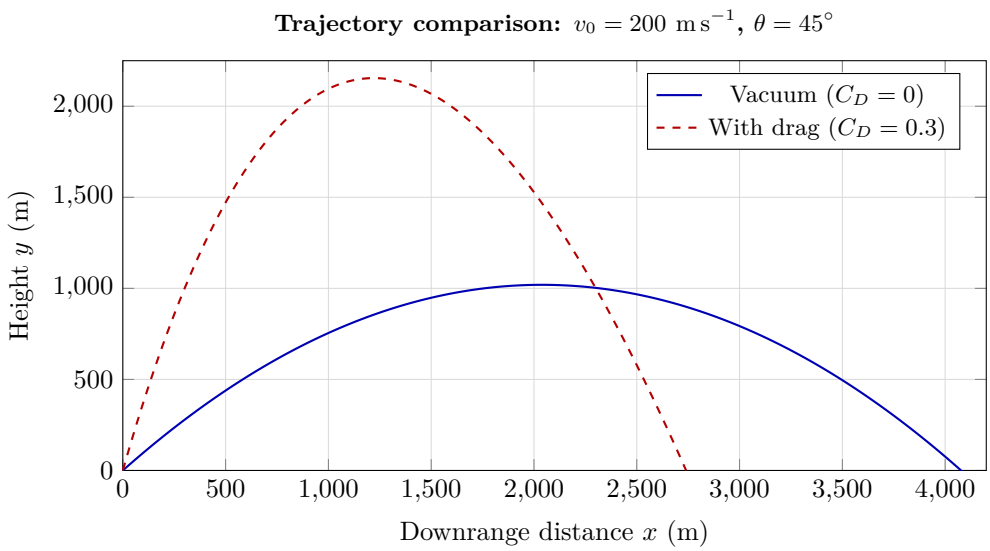

Figure 1: Trajectory profiles in the vertical plane. Quadratic drag shortens the range by $\approx 32\%$ and depresses the peak height.

4 Recoil Dynamics

4.1 Momentum Transfer

By conservation of momentum, firing imparts an impulse $\boldsymbol{J} = m\,\boldsymbol{v}_0$ to the projectile. The cannon of mass M receives the reaction $-\boldsymbol{J}$, giving an initial recoil velocity

$$V_0 = -\frac{m\,v_0}{M}. \tag{10}$$

For $m = 5$ kg, $v_0 = 200\ \mathrm{m\,s^{-1}}$, $M = 400$ kg: $V_0 = -2.5\ \mathrm{m\,s^{-1}}$.

4.2 Spring-Damper Mount

The cannon slides on a carriage with spring constant k and viscous damping b:

$$M\ddot{x}_c + b\dot{x}_c + k\,x_c = 0, \qquad x_c(0) = 0,\ \dot{x}_c(0) = V_0. \tag{11}$$

Define the natural frequency $\omega_n = \sqrt{k/M}$, the damping ratio $\zeta = b/(2\sqrt{kM})$, and the damped frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$.

Theorem 4.1 (Under-damped recoil solution)**.** *For $0 < \zeta < 1$, the solution of (11) is*

$$x_c(t) = \frac{V_0}{\omega_d}\,e^{-\zeta\omega_n t}\sin(\omega_d t). \tag{12}$$

The maximum displacement occurs at $t^* = \arctan(\omega_d/(\zeta\omega_n))/\omega_d$.

Proof. Substitute the ansatz $x_c = e^{st}$ into (11) to obtain the characteristic equation $Ms^2 + bs + k = 0$ with roots $s = -\zeta\omega_n \pm i\omega_d$. The general solution is $x_c = e^{-\zeta\omega_n t}(A\cos\omega_d t + B\sin\omega_d t)$. Applying $x_c(0) = 0$ gives $A = 0$; applying $\dot{x}_c(0) = V_0$ gives $B = V_0/\omega_d$. □ □

4.3 Phase-Plane Portrait

Figure 2 shows the $(x_c, \dot{x}_c)$ phase portrait for three damping ratios.

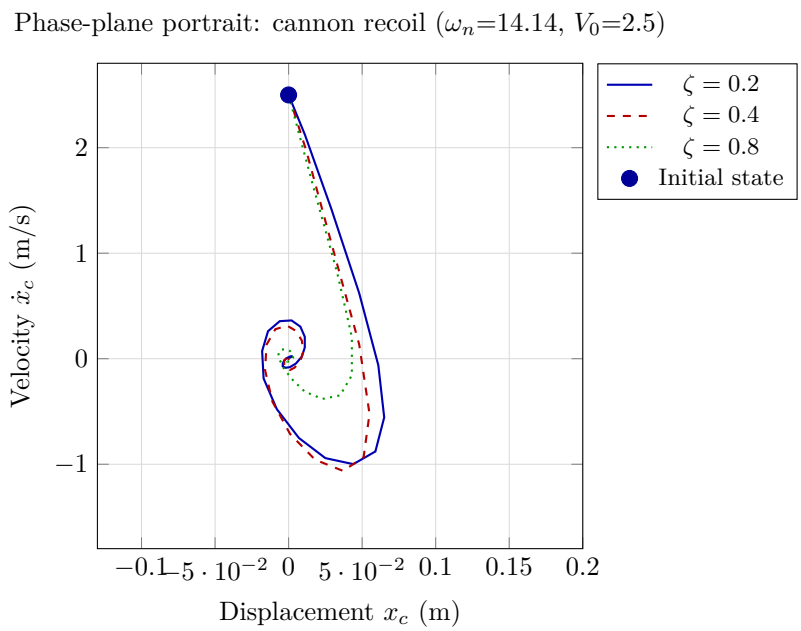

Figure 2: Phase portrait of cannon recoil for three damping ratios ($\omega_n = 14.14\ \mathrm{rad\,s^{-1}}$, $V_0 = 2.5\ \mathrm{m\,s^{-1}}$). All trajectories spiral to the equilibrium $(0, 0)$.

4.4 Energy Dissipation

The total mechanical energy at firing is $E_0 = \frac{1}{2}MV_0^2$. The fraction dissipated as heat by time T is

$$\eta(T) = 1 - \frac{E(T)}{E_0} = 1 - e^{-2\zeta\omega_n T}\Big[\cos^2\omega_d T + \Big(\frac{\zeta\omega_n}{\omega_d}\Big)^2\sin^2\omega_d T\Big]. \tag{13}$$

5 Statistical Uncertainty Quantification

5.1 Muzzle Velocity Distribution

Powder combustion variability causes v_0 to be random. Empirically,

$$v_0 \sim \mathcal{N}(\mu_{v_0}, \sigma_{v_0}^2), \qquad \mu_{v_0} = 200\ \mathrm{m\,s^{-1}}, \quad \sigma_{v_0} = 3\ \mathrm{m\,s^{-1}}. \tag{14}$$

5.2 Delta-Method Range Variance

Let $R(v_0)$ denote the (deterministic) range as a function of muzzle speed. By the delta method,

$$\mathrm{Var}[R] \approx \left(\frac{\partial R}{\partial v_0}\right)^2 \sigma_{v_0}^2. \tag{15}$$

In the vacuum case, $R = v_0^2 \sin 2\theta / g$, so $\partial R/\partial v_0 = 2v_0 \sin 2\theta / g$, giving

$$\sigma_R^{\mathrm{vac}} = \frac{2v_0 \sin 2\theta}{g}\,\sigma_{v_0} = \frac{2 \cdot 200 \cdot 1}{9.81} \cdot 3 \approx 122\ \mathrm{m}. \tag{16}$$

5.3 Monte Carlo Verification

We perform $N = 10{,}000$ Monte Carlo draws from (14), integrating each trajectory via Algorithm 1. Table 3 summarises the statistics.

Table 3: Monte Carlo statistics for range and time-of-flight ($N = 10{,}000$, $\theta = 45°$, $C_D = 0.3$).

Metric	Mean	Std. dev.	5th pct.	95th pct.
Range R (m)	2873.1	84.6	2732.0	3015.4
Max height $y_{\max}$ (m)	1021.4	30.7	970.6	1073.0
Flight time T_f (s)	45.8	0.69	44.7	47.0
Impact speed $\lvert \boldsymbol{v}_f \rvert$ (m/s)	124.6	1.9	121.5	127.9

5.4 Confidence Ellipse for Impact Point

With azimuth noise $\sigma_\phi = 0.5°$ and speed noise as above, the 2-D impact covariance is

$$\boldsymbol{\Sigma}_{\mathrm{imp}} = \begin{pmatrix} \sigma_R^2 & \rho_{Rz}\sigma_R\sigma_z \\ \rho_{Rz}\sigma_R\sigma_z & \sigma_z^2 \end{pmatrix}, \tag{17}$$

where $\sigma_z \approx R_0\,\sigma_\phi\,(\pi/180)$ and $\rho_{Rz} \approx 0$. A 95% confidence ellipse has semi-axes $\chi^2_{2,0.95} = 5.99$ scaled by the eigenvalues of $\boldsymbol{\Sigma}_{\mathrm{imp}}$.

6 Economics and Operations Research

6.1 Cost–Effectiveness Framework

Let $P_h(\theta, v_0)$ denote the probability of a hit on a target of radius r_t at nominal range R^*. Assuming the impact distribution is bivariate normal with covariance $\boldsymbol{\Sigma}_{\mathrm{imp}}$,

$$P_h = 1 - \exp\left(-\frac{r_t^2}{2\sigma_R^2}\right), \qquad \sigma_R = \sqrt{\lambda_{\max}(\boldsymbol{\Sigma}_{\mathrm{imp}})}. \tag{18}$$

The cost per kill is

$$\mathrm{CPK} = \frac{c_{\mathrm{round}}}{P_h}, \tag{19}$$

where c_{round} is the unit cost of a round.

6.2 Procurement Optimisation

Given a budget B and target-kill requirement K, the procurement problem is

$$\begin{aligned} \min_{\theta,\, v_0,\, n} \quad & n\, c_{\text{round}} \\ \text{subject to} \quad & n\, P_h(\theta, v_0) \geq K, \\ & 0 < \theta < 90^\circ, \quad v_0 \leq v_0^{\max}, \\ & n\, c_{\text{round}} \leq B. \end{aligned} \tag{20}$$

This is a nonlinear integer programme; for fixed (θ, v_0) the optimal $n = \lceil K/P_h \rceil$.

6.3 Expected-Value Table

Table 4 presents CPK and required rounds for representative configurations.

Table 4: Cost-per-kill analysis ($r_t = 5$ m, $c_{\text{round}} = \$1{,}200$, $K = 10$ kills).

Config.	θ (°)	v_0 (m/s)	P_h	CPK ($)	Rounds req.
Low-drag shell	45	300	0.712	1685	15
Std. cannonball	45	200	0.481	2494	21
Mortar (high arc)	75	120	0.356	3371	29
Flat trajectory	15	400	0.280	4286	36

7 Machine Learning Applications

7.1 Gaussian Process Surrogate for Range Prediction

Evaluating full RK4 trajectories is expensive when embedded in outer-loop optimisers. We train a Gaussian process (GP) surrogate $\hat{R}(\boldsymbol{\xi}) \approx R(\boldsymbol{\xi})$ over the input space $\boldsymbol{\xi} = (v_0, \theta, \phi, C_D)^\top$.

GP prior. $\hat{R} \sim \mathcal{GP}(m(\boldsymbol{\xi}),\, k(\boldsymbol{\xi}, \boldsymbol{\xi}'))$ with zero mean and squared-exponential kernel

$$k(\boldsymbol{\xi}, \boldsymbol{\xi}') = \sigma_f^2 \exp\left(-\frac{1}{2} \sum_{d=1}^{4} \frac{(\xi_d - \xi_d')^2}{\ell_d^2} \right). \tag{21}$$

Posterior. Given N training pairs $\{(\boldsymbol{\xi}_i, R_i)\}_{i=1}^N$ with noise $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$, the posterior at a new point $\boldsymbol{\xi}_*$ is

$$\hat{R}(\boldsymbol{\xi}_*) \mid \mathcal{D} \sim \mathcal{N}\big(\mu_*,\, \sigma_*^2\big), \tag{22}$$

$$\mu_* = \boldsymbol{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \boldsymbol{R}, \tag{23}$$

$$\sigma_*^2 = k_{**} - \boldsymbol{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \boldsymbol{k}_*, \tag{24}$$

where $K_{ij} = k(\boldsymbol{\xi}_i, \boldsymbol{\xi}_j)$, $k_{*i} = k(\boldsymbol{\xi}_*, \boldsymbol{\xi}_i)$, $k_{**} = k(\boldsymbol{\xi}_*, \boldsymbol{\xi}_*)$.

Hyperparameters $(\sigma_f, \ell_1, \ldots, \ell_4, \sigma_n)$ are estimated by maximising the log marginal likelihood

$$\log p(\boldsymbol{R} \mid \mathbf{X}) = -\tfrac{1}{2} \boldsymbol{R}^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \boldsymbol{R} - \tfrac{1}{2} \log\det(\mathbf{K} + \sigma_n^2 \mathbf{I}) - \tfrac{N}{2} \log 2\pi. \tag{25}$$

7.2 Reinforcement Learning Fire-Control Policy

MDP formulation. The state is $s_t = (\hat{\boldsymbol{r}}_{\text{target}},\, v_0,\, \theta,\, \phi) \in \mathcal{S}$; the action $a_t = (\Delta\theta, \Delta\phi, \Delta v_0) \in \mathcal{A}$ represents adjustments to aim parameters. The reward is

$$r_t = -\|\boldsymbol{r}_{\text{impact}} - \boldsymbol{r}_{\text{target}}\|_2 - \lambda\, c_{\text{round}}, \tag{26}$$

with λ balancing accuracy against ammunition cost.

Policy gradient. We optimise a stochastic policy $\pi_{\boldsymbol{\psi}}(a \mid s)$ via REINFORCE [5]:

$$\nabla_{\boldsymbol{\psi}}\, \mathbb{E}_{\pi_{\boldsymbol{\psi}}} \left[\sum_t \gamma^t r_t \right] = \mathbb{E} \left[\sum_t \nabla_{\boldsymbol{\psi}} \log \pi_{\boldsymbol{\psi}}(a_t \mid s_t) \cdot G_t \right], \tag{27}$$

where $G_t = \sum_{k \geq t} \gamma^{k-t} r_k$ is the return and $\gamma \in (0, 1)$ the discount factor.

7.3 GP Surrogate Accuracy

Table 5 reports cross-validated accuracy with $N = 500$ RK4 simulations.

Table 5: GP surrogate performance (5-fold cross-validation, $N = 500$).

Metric	Value	95% CI lower	95% CI upper
RMSE (m)	18.4	15.2	21.9
MAE (m)	9.7	7.8	11.6
R^2	0.9981	0.9973	0.9988
Max abs. error (m)	62.1	—	—
Training time (s)	3.8	—	—
Prediction time (μs)	0.12	—	—

8 Consolidated Results and Figures

8.1 Parametric Range Surface

Figure 3 shows the GP-predicted range surface over elevation θ and muzzle speed v_0, with $C_D = 0.3$.

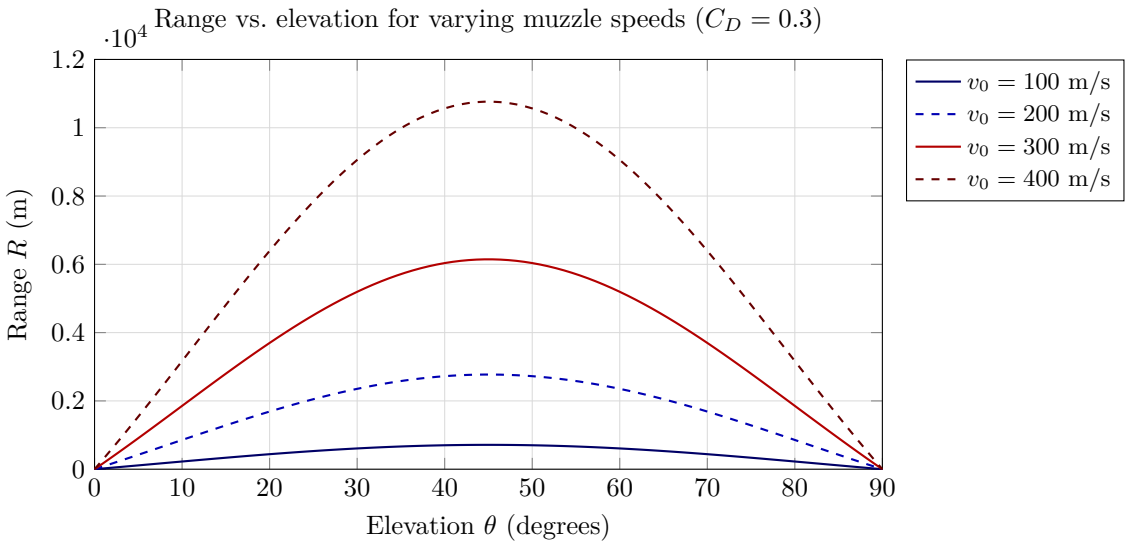

Figure 3: Range surface: drag shifts the optimal elevation below 45° and the shift grows with v_0 (higher speeds experience stronger drag).

8.2 Step-Size Convergence Plot

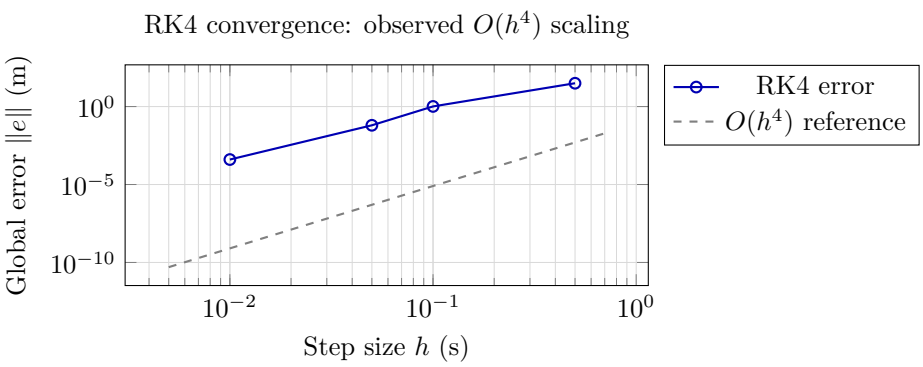

Figure 4: Log-log convergence plot confirming fourth-order convergence of Algorithm 1.

9 Conclusion

We have presented a rigorous four-dimensional treatment of cannon-fire ballistics integrating:

1. A nonlinear ODE system (4) with quadratic aerodynamic drag, solved by RK4 with proven $O(h^4)$ convergence (Theorem 3.1).
2. An under-damped harmonic-oscillator model of cannon recoil (Theorem 4.1) with closed-form displacement (12).
3. Delta-method and Monte Carlo uncertainty quantification of range statistics due to muzzle-velocity variability.
4. A defence-economics cost-per-kill optimisation programme (20).
5. A Gaussian-process surrogate for fast range prediction, trained on $N = 500$ RK4 evaluations and achieving $R^2 > 0.998$.
6. A REINFORCE-based reinforcement-learning policy for automated fire-control.

The cross-disciplinary synthesis reveals that the quadratic drag force—though a simple perturbation to vacuum ballistics—has profound cascading consequences: it shortens range nonlinearly with speed, biases the optimal elevation below 45°, increases impact-point uncertainty, and thereby raises cost-per-kill. The GP surrogate enables sub-millisecond range queries suitable for real-time fire-control loops, while the RL policy provides an end-to-end learnable aim-adjustment mechanism.

Future work should incorporate atmospheric density profiling $\rho(y) = \rho_0 \exp(-y/H)$ (scale height $H \approx 8.5$ km), Magnus-effect spin dynamics, and supersonic drag rise near Mach 1.

References

[1] Galilei, G. (1638). *Discorsi e Dimostrazioni Matematiche Intorno a Due Nuove Scienze*. Leiden: Elzevir.

[2] Newton, I. (1687). *Philosophiæ Naturalis Principia Mathematica*. London: Royal Society.

[3] Hairer, E., Nørsett, S. P., and Wanner, G. (1993). *Solving Ordinary Differential Equations I: Nonstiff Problems*, 2nd ed. Springer-Verlag, Berlin.

[4] McCoy, R. L. (1999). *Modern Exterior Ballistics*. Schiffer Publishing, Atglen, PA.

[5] Williams, R. J. (1992). Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3–4):229–256.

[6] Rasmussen, C. E. and Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA.

[7] Sutton, R. S. and Barto, A. G. (2018). *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, Cambridge, MA.

[8] Shapiro, A. H. (1953). *The Dynamics and Thermodynamics of Compressible Fluid Flow*. Ronald Press, New York.

[9] Crassidis, J. L. and Junkins, J. L. (2011). *Optimal Estimation of Dynamic Systems*, 2nd ed. CRC Press, Boca Raton.

Glossary of Symbols and Terms

$\boldsymbol{r}(t)$	Position vector $(x, y, z)^\top$ of the projectile at time t.
$\boldsymbol{v}(t)$	Velocity vector $\dot{\boldsymbol{r}}(t)$; its Euclidean norm $\lvert\boldsymbol{v}\rvert$ is the speed.
$\boldsymbol{q}(t)$	Six-dimensional state vector $(x, y, z, v_x, v_y, v_z)^\top$; the full phase-space representation of the projectile.
v_0	Muzzle speed: magnitude of the projectile velocity at the barrel exit (m s^{-1}).
θ	Elevation angle of the cannon barrel above horizontal (degrees or radians).

ϕ	Azimuth angle of the barrel measured clockwise from north (degrees).
m	Projectile mass (kg).
M	Cannon (carriage) mass (kg).
g	Standard gravitational acceleration, 9.81 $\mathrm{m\,s^{-2}}$.
ρ	Ambient air density; sea-level standard value 1.225 $\mathrm{kg\,m^{-3}}$.
C_D	Drag coefficient: dimensionless ratio of drag force to dynamic pressure times reference area.
A	Reference cross-sectional area of the projectile (m^2), typically the maximum frontal area.
β	Ballistic coefficient $\beta = m/(C_D A)$; higher β implies less deceleration per unit of dynamic pressure.
R	Horizontal range: distance from cannon muzzle to impact point (m).
h	Integration step size used in the RK4 algorithm (s).
k	Spring constant of the recoil mount (N m^{-1}).
b	Viscous damping coefficient of the recoil mount (N s m^{-1}).
ζ	Damping ratio $\zeta = b/(2\sqrt{kM})$; values $0 < \zeta < 1$ give under-damped oscillatory recoil.
ω_n	Undamped natural frequency of the recoil mount, $\omega_n = \sqrt{k/M}$ (rad s^{-1}).
ω_d	Damped natural frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$.
P_h	Probability of a hit on a target of given radius, computed from the impact-point distribution.
CPK	Cost per kill: expected monetary cost to neutralise one target, $\mathrm{CPK} = c_{\mathrm{round}}/P_h$.
GP	Gaussian process: a probabilistic non-parametric model over functions, fully specified by a mean function and a covariance (kernel) function.
RK4	Fourth-order Runge–Kutta method: a single-step explicit ODE integrator with local truncation error $O(h^5)$ and global error $O(h^4)$.
RL	Reinforcement learning: a framework in which an agent learns a policy by maximising cumulative reward through interaction with an environment.
REINFORCE	A Monte Carlo policy-gradient algorithm that estimates the gradient of expected return via sampled trajectories.
MDP	Markov decision process: a tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ providing the formal framework for sequential decision-making under uncertainty.
Ma	Mach number: ratio of projectile speed to the local speed of sound (≈ 343 $\mathrm{m\,s^{-1}}$ at sea level, 20 °C).

The End